The L9-Cognitive Graph: Nine Layers from Lexical Decomposition to Paradigm Generation

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> *"A graph that answers `[Apple] --is_a--> [Fruit]` is a dictionary. Maxwell did not consult a dictionary."*

Abstract

Probabilistic next-symbol prediction confines a system to interpolating within existing knowledge. We propose the **L9-Cognitive Graph**, a neuro-symbolic architecture organised as nine discrete layers of traversal logic over a deterministic symbolic graph, ascending from lexical decomposition to the generation of new axiomatic roots.

The organising claim is that breakthrough reasoning is not semantic retrieval but **cross-domain isomorphism**: recognising that the mathematical structure of one domain describes the

unresolved anomalies of another. Achieving this requires the engine to move from *constraint enforcement* to *abductive constraint generation.*

This paper specifies all nine layers — each with its input, operation, output and failure mode — rather than listing them. Section 5 states, layer by layer, which are implemented, which are partial, and which remain proposals. Four are live, two are partial, and three are not built.

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1. The Limits of Semantic Retrieval

A graph database mapping `[Apple] --is_a--> [Fruit]` is a semantic dictionary: it returns what was stored, structured. Human breakthrough logic does not work this way. Maxwell's unification of electricity and magnetism was not a lookup; it was the recognition that one domain's mathematical structure resolved another's anomalies.

Reproducing that requires two capabilities a dictionary lacks. The engine must be able to notice that its own knowledge is **structurally inconsistent** rather than merely incomplete, and it must be able to **propose structure that was never asserted** in response.

The layers below are ordered by how far each departs from retrieval.

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2. Biological Motivation: Two Topologies

The architecture borrows two structures rather than a metaphor.

Neocortical columns — the local graph. The brain processes localised, highly deterministic rules: syntax, basic physics. These correspond to densely connected subgraph clusters with near-unit edge weights, where traversal is reliable and cheap.

The Default Mode Network — the abductive engine. The DMN activates during undirected thought and deep problem-solving, searching for low-probability, long-distance connections across disparate columns. It corresponds to an asynchronous background process running bridge-finding across structurally similar but semantically distant subgraphs.

The functional distinction that matters: **the local graph answers, the DMN wanders.** A system with only the first can never surprise itself.

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3. The Nine Layers

Each layer is specified by what it takes, what it does, what it emits, and how it fails. Layers are cumulative: Ln presupposes L1..n−1.

L1 — Lexical / Morphological Decomposition

Input. A token that resolves to no node. **Operation.** Fracture the string into sub-lexical fragments — morphemes, syllables, character clusters — and project each against the ontology for partial matches. `biology` decomposes to `bio` + `logy`; `sudo` to `su` + `do`. **Output.** A set of candidate bindings for an otherwise orphan node. **Failure mode.** Spurious decomposition. Short fragments match everything, so a fragment carries usable evidence only above a minimum length — an information floor, not a stopword list.

This layer exists because a node of degree zero is unreachable, and an unreachable node makes every higher layer inapplicable.

L2 — Ontological Retrieval

Input. A resolved node. **Operation.** Traverse explicit edges. `Darwin -> Human`. **Output.** Asserted neighbours. **Failure mode.** Silence when no edge exists. This is the honest failure and is preferable to a fabricated one.

L1 and L2 together constitute the dictionary. Everything above is the argument of this paper.

L3 — Transitive Inference

Input. Two chained edges $A \to B$, $B \to C$. **Operation.** Instantiate the implied edge $A \to C$, recording both premises. **Output.** A derived edge, distinguishable from an asserted one. **Failure mode.** Unbounded expansion. Transitive closure over a large graph is quadratic, so derivation must be bounded by an explicit budget.

The provenance requirement is not decoration. A derived edge that cannot name its premises cannot be retracted when one of them is withdrawn.

L4 — Causal Abduction

Input. An observed end state Y. **Operation.** Search *backwards* for a cause X whose consequences include Y. This inverts L3's direction of travel. **Output.** A hypothesis, not a conclusion. **Failure mode.** Affirming the consequent. Many causes produce the same effect, so an abduced X is a candidate requiring independent support — and labelling it otherwise is the single easiest way for this architecture to produce confident nonsense.

L5 — Counterfactual Simulation

Input. A core axiom. **Operation.** Temporarily invert it — *what if gravity repelled?* — and propagate consequences, checking whether the resulting system remains consistent. **Output.** A stability verdict on the inverted world. **Failure mode.** Combinatorial explosion, and contamination of the base graph if the inversion is not perfectly isolated and reverted.

L5 is how the engine determines which axioms are *load-bearing*: an axiom whose inversion changes nothing was doing no work.

L6 — Analogical Mapping

Input. Two subgraphs from different domains. **Operation.** Attempt a structure-preserving 1:1 correspondence — fluid dynamics onto electrical circuits, pressure $\to$ voltage, flow $\to$ current. **Output.** A mapping, plus the edges of one domain transferred onto the other as hypotheses. **Failure mode.** Superficial isomorphism. Two subgraphs can share shape while the correspondence means nothing, so a mapping must be scored by more than topology.

This is the first layer of genuine novelty, because its output is not entailed by either domain alone.

L7 — Anomaly Detection

Input. The graph. **Operation.** Scan for nodes of high connectivity carrying *contradictory* constraints — a concept that participates widely while holding incompatible commitments. This is the **Maxwell gap**: the signature of a theory that is incomplete at a specific, locatable point. **Output.** A ranked list of structural contradictions. **Failure mode.** Mistaking noise for anomaly. Extraction error produces contradictions that indicate a bad parse rather than a gap in the world.

L7 is the layer that lets the system be *dissatisfied* with what it knows, which is the precondition for L8 and L9.

L8 — Synthesis (The DMN Layer)

Input. An L7 anomaly. **Operation.** Generate a **bridge node** resolving it, constrained to satisfy both of the subgraphs it joins. **Output.** A new vertex that appeared in no source. **Failure mode.** Unfalsifiable patching. A bridge invented purely to absorb a contradiction, with no independent consequences, explains nothing — it renames the problem.

The discipline required here is that a synthesised node must make predictions beyond the anomaly that motivated it.

L9 — Paradigm Generation

Input. A region of the graph that is verbose, irregular and patched. **Operation.** Define a new **axiomatic root** from which the messy region follows, collapsing many special cases into fewer general ones. **Output.** A reorganisation of the knowledge base, not an addition to it. **Failure mode.** Losing content. A simplification that fails to entail what it replaced is not a paradigm; it is deletion.

L9 is the only layer whose success criterion is *compression*: the same commitments held in less structure.

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4. Scoring Novelty

Novelty is not randomness — it is the discovery of hidden structural isomorphism. Given subgraphs A and B, we seek pairs that are **semantically distant** but **structurally identical**:

$$\Phi(A, B) \;=\; \frac{I(A, B)}{e^{-k \cdot d(A,B)}}$$

where $I(A, B)$ is a maximal common subgraph isomorphism score, $d(A, B)$ is the shortest path between the subgraph roots, and k is a scaling constant.

The exponential in the denominator is the load-bearing choice: it makes Φ grow with distance, so a structural match between *adjacent* concepts scores poorly. Two neighbouring nodes sharing shape is expected. Two nodes twelve hops apart sharing shape is a candidate breakthrough. Pairs exceeding a threshold τ are flagged.

A caution this formulation invites. Φ rewards distance unboundedly, so with a finite isomorphism score the highest-scoring pairs will tend to be the most remote — which includes pairs that are remote because one of them is extraction noise. Any deployment needs a floor on subgraph quality before distance is rewarded.

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5. Implementation Status

The paper's claims are separated from the engine's behaviour here, layer by layer.

Layer	Status	Where, or what is missing
L1 Morphological	**live**	bigram sensory lanes; graded recognition survives misspelling, 7/7 measured
L2 Retrieval	**live**	typed concept graph traversal
L3 Transitive	**live**	forward chaining with premise citation; derived edges name both parents
L4 Abduction	**partial**	abduction over upstream cavities exists; no general backward search from an arbitrary Y
L5 Counterfactual	**not built**	requires isolated graph forking with guaranteed revert; no such isolation exists
L6 Analogical	**live**	shared-target intersection plus rubber-tunnel bridging, with 20-seed reproducibility data; the Fisher metric supplies a principled distance
L7 Anomaly	**partial**	contradiction detection and isolated-subgraph/logic-gap scans exist; no ranked Maxwell-gap register
L8 Synthesis	**partial**	fission spawns new vertices to resolve contradiction, and mutation forces distant-pair hypotheses that decay if unsupported — but neither is *driven by* an L7 anomaly report
L9 Paradigm	**not built**	no mechanism reorganises the graph around a new root; macro-rule induction compresses *derivations*, not *axioms*, which is a weaker relative

The honest summary. The lower half of the stack is real. The upper half is partially present but *disconnected*: the engine has anomaly detection and it has synthesis, but synthesis does not consume anomaly reports. Wiring L7 into L8 is the single highest-value unbuilt link, and it is engineering rather than research.

L5 and L9 are genuinely open. L9 in particular is not a matter of effort: no criterion is currently specified for when a proposed axiomatic root is *better* than the structure it would replace, and without that criterion the layer cannot be evaluated even if built.

6. Limitations

The layer ordering is asserted. Nine layers with clean boundaries is a presentational convenience. In the running engine, L1 recognition and L6 bridging interact directly, and L3 derivation feeds L7 detection without passing through L4 or L5. The stack is a taxonomy, not a pipeline.

Φ is untuned. The constant k and threshold τ have no principled setting, and §4 notes the formulation's bias toward remote noise. The Fisher-information distance is a more defensible replacement for d, and supersedes the shortest-path form where both are available.

No layer above L3 has an oracle. L3's output is checkable by construction. An L6 mapping, an L8 bridge or an L9 paradigm has no external verdict available — which is precisely the gap the compile-verify loop closes for code, and does not close for physics or biology.

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7. Conclusion

The L9 stack describes the ascent from a semantic dictionary to an isomorphic constraint-satisfaction engine — one able to observe structural contradictions in what it holds and calculate the structure required to unify them.

Four layers are live and two are partial. The most useful outcome of writing the stack out in full is not the top of it but the middle: it makes visible that anomaly detection and synthesis both exist and are **not connected to each other**, which was not apparent while the layers were a list of one-line descriptions.

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